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Website: https://simonwan03.github.io

Education

- **Shanghai Jiao Tong University** Shanghai, China
  Master's Degree in Computer Science *2026 – 2028 (Expected)*
  Dual Degree Program with École Polytechnique
- **École Polytechnique** Paris, France
  Diplôme d'Ingénieur (Master's Level) *2024 – 2027 (Expected)*
  Major in Artificial Intelligence
- **Shanghai Jiao Tong University** Shanghai, China
  B.Eng. in Information Engineering & B.A. in French Literature *2022 – 2026*

Research

- **Benchmarking Machine Learning for Chaotic Three-Body Dynamics in Astrophysics** 2026
  ***Sheng Wan**, Pavan Vynatheya, Duo Xu, Kevin Li, Ye Zhu*
  NeurIPS 2026 Workshop: Sim2Science: ML with Imperfect Scientific Models

Experience

- **Shanghai Luoshu Investment** Shanghai, China
  Quantitative Research Intern *June 2026 – August 2026*
  - **Volatility Modeling & Systematic Options Trading**: Built an end-to-end minute-level volatility research pipeline for commodity futures options, including IV calibration, Greeks estimation, and volatility surface construction. Developed and backtested delta-hedged straddle strategies driven by IV–RV signals.
  - **Treasury Cash-Futures Relative Value**: Developed a Treasury cash-futures relative value framework that maps Chinese government bond futures to cash bond fair values through CTD, conversion factor, basis, and implied-repo analytics; constructed duration/DV01-neutral relative-value signals across TS/TF/T/TL and cash bonds to capture short-term pricing dislocations.
- **LISN, Université Paris-Saclay** Saclay, France
  Research Intern *July 2025 – August 2025*
  - **ANaGRAM for PINNs Optimization**: Applied ANaGRAM, an adaptive natural-gradient method based on the Neural Tangent Kernel (NTK), to improve the convergence speed and accuracy of Physics-Informed Neural Networks (PINNs).
  - **Adaptive Collocation Point Selection**: Developed a collocation-embedding-guided adaptive sampling strategy for ANaGRAM, improving collocation point selection and achieving faster convergence and higher accuracy during NTK-based projection.
- **A.O.Smith** Nanjing, China
  Software Developer Intern *July 2023 – August 2023*
  - **IoT Data Platform Development**: Built a real-time MQTT-based IoT data pipeline using Java Spring, enabling configurable data processing through reflection and ensuring concurrency-safe operations with Redis distributed locks.
  - **Database Optimization**: Resolved SQL pagination issues and optimized database queries, improving data retrieval reliability and query performance.

Awards

- **China Collegiate Programming Contest (CCPC), Shanghai Region, 2022**: *Silver Prize*
- **National Olympiad in Mathematics in Provinces, Jiangsu, 2021**: *Second Prize*
- **National Olympiad in Physics in Provinces, Jiangsu, 2021**: *Second Prize*
- **National Olympiad in Informatics in Provinces (NOIP), Jiangsu, 2020**: *First Prize*

Skills

- **Languages**: Chinese (Native), English (C1), French (C1)
- **Programming**: Python, C++, Java, SQL
- **ML & Data**: PyTorch, JAX, NumPy, pandas, scikit-learn
- **Tools**: Git, Linux, LaTeX